An *Ab Initio* Rovibrational Linelist for HPO

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Motivation: Why HPO?

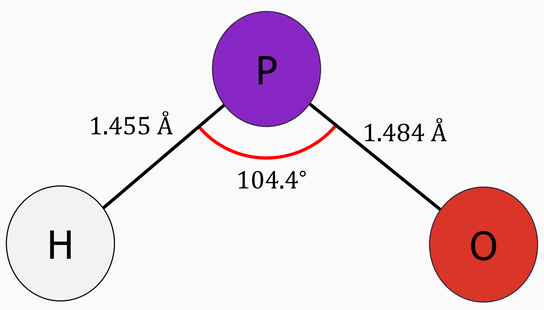
HPO appears as a precursor molecule in proposed abiotic mechanisms to form phosphine on Venus[1]. It has been used for modeling the atmosphere of 55 Cancri e and predicted to be detectable in the interstellar medium[2,3].

Can HPO play a role on lava or rocky exoplanets?

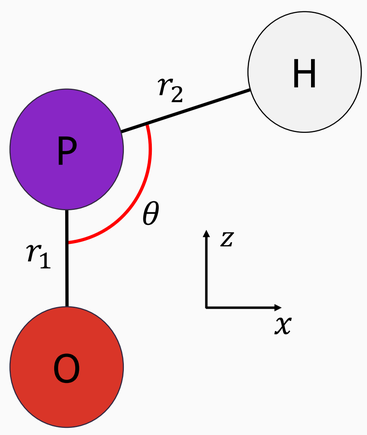
Aims

- Produce a rovibrational linelist for HPO (electronic ground state only).
- Evaluate potential for HPO detection on exoplanets/planets.

Equilibrium bond lengths and angles at the electronic ground state ($\tilde{X}\ ^1A'$)

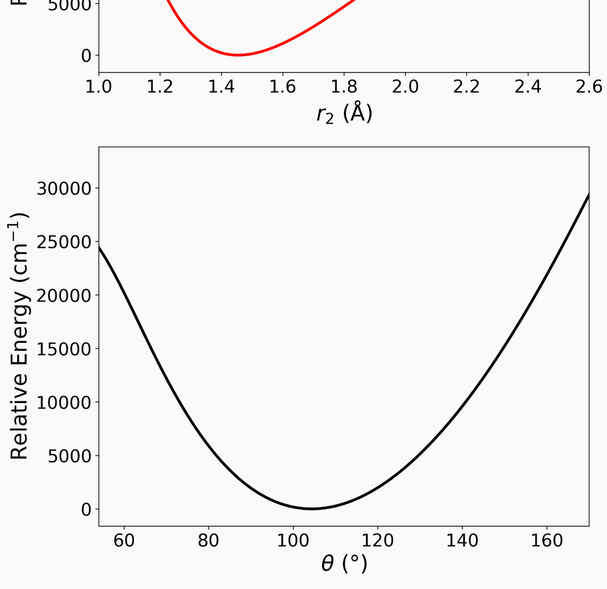


We use the internal coordinates and Cartesian axis as presented:



Potential Energy Surface

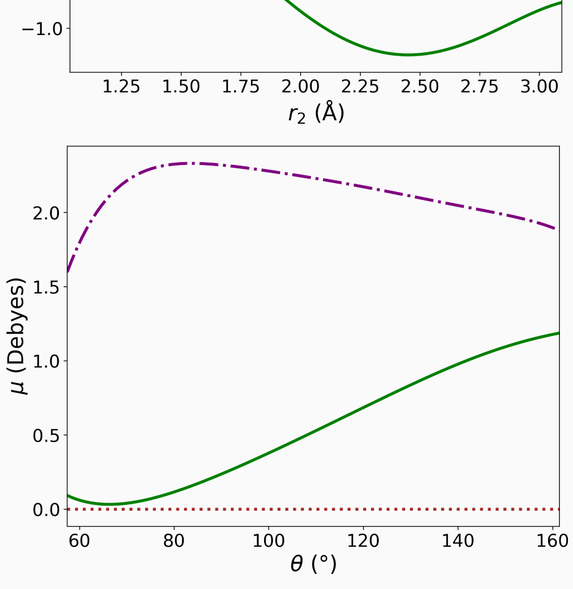
7770 *ab initio* points computed with CCSD(T)-F12/cc-pVQZ-F12a



- Fit PES has root-mean-square error of 2 cm^{-1} for energies under 18,000 cm^{-1} .
- Molecule appears relatively rigid (stiff bending), we can use non-singular methods to perform variational calculations.

Dipole Moment Surfaces

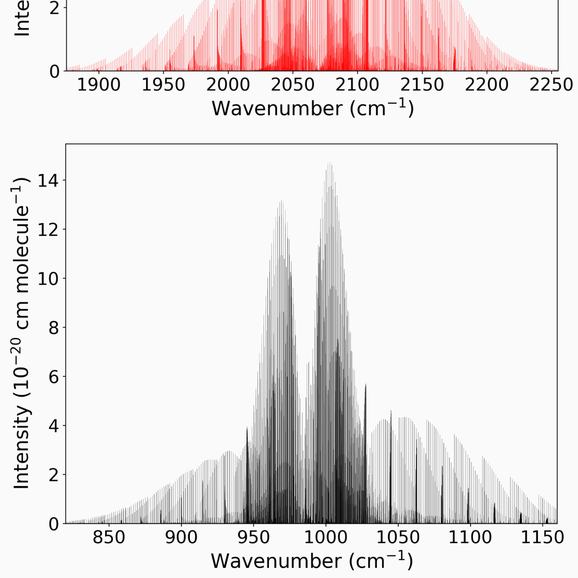
Same grid as the PES computed with CCSD(T)/aug-cc-pV(5+d)Z



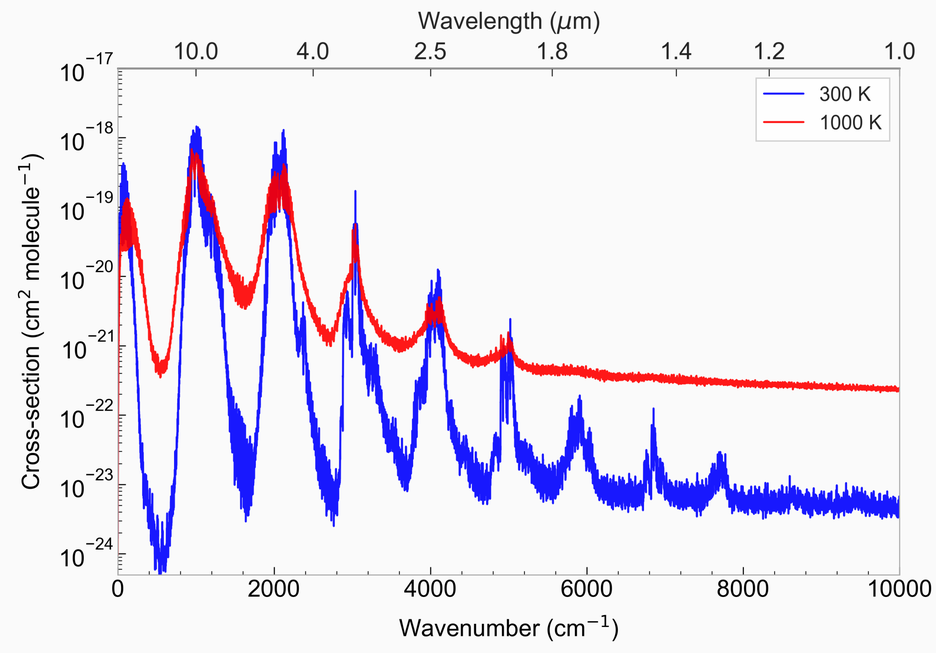
- Dipole values calculated oriented with respect to the Cartesian axis given.
- Fitted to a Taylor-series type expansion. Dipole values along the y-axis are 0 since the molecule is planar.

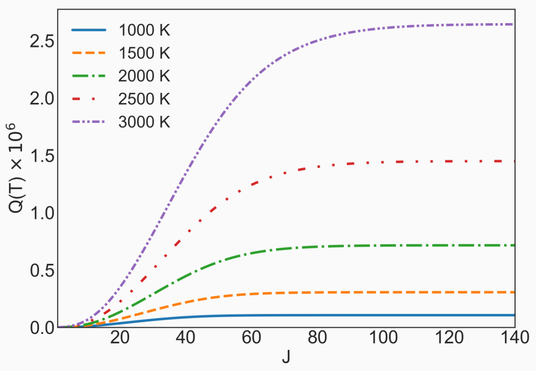
Stick Spectra

Rotational stick spectra of the fundamental vibrational modes



- The first and third fundamental modes have been experimentally measured, but not the second, despite intensity seeming higher.
- Experimental values can be used to refine rovibrational calculations.





Cross-sections computed with states up to J = 141. Partition function plots as a function of J suggests J = 120 could be sufficient instead.

Cross-sections and partition functions were computed with the program ExoCross[5].

Cross-sections

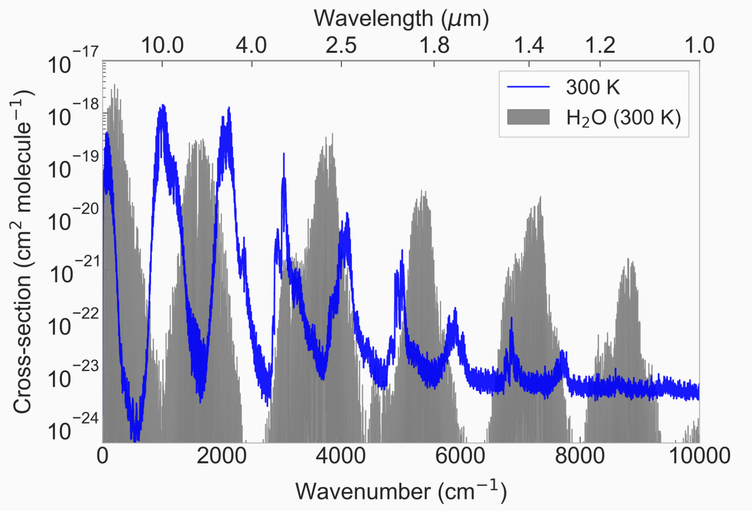
Rovibrational energies and wavefunctions are computed with variational nuclear motion program TROVE[4], then used to compute stick spectra and cross sections.

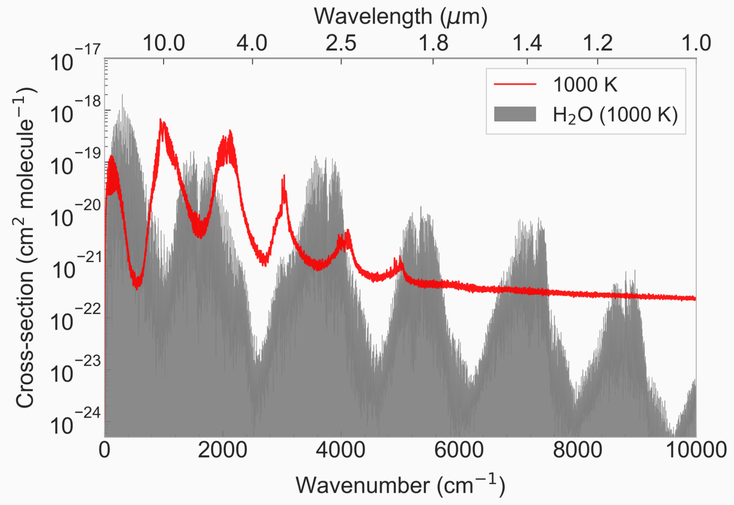
Lower states are selected between 0 to 14,000 cm^{-1} while upper states are between 0 to 24,000 cm^{-1}. Includes 2.4 million rovibrational states and 2.2 x 10^{10} transitions.

Exoplanet Detection?

The fundamental bands of HPO do not overlap with water. This makes detection more feasible. The fundamental band at 985 cm^{-1} would be particularly visible.

We suggest HPO can be of interest for studying exoplanets and the planetary chemistry of phosphorus. A line list for HPO should be provided on the ExoMol website soon.





References:
1. K. Mráziková, A. Knížek, H. Saeidfirozeh, et al., 'A Novel Abiotic Pathway for Phosphine Synthesis over Acidic Dust in Venus' Atmosphere.', *Astrobiology*. 2024;24(4):407–422. doi:10.1089/ast.2023.0046
2. M. Zilinskas, C. P. A. van Buchem, S. Zieba, Y. Miguel, E. Sandford, R. Hu, J. A. Patel, A. Bello-Arufe, L. J. Janssen, S.-M. Tsai, D. Dragomir, M. Zhang, 'Characterising the atmosphere of 55 Cancri e – 1D forward model grid for current and future JWST observations', A&A 697 A34 (2025), doi:10.1051/0004-6361/202554062
3. B. E. Turner, T. Tsuji, J. Bally, M. Guélin, J. Cernicharo, 'Phosphorus in the Dense Interstellar Medium'. *Astrophys. J.* 1990, *365*, 569. https://doi.org/10.1086/169511
4. S. N. Yurchenko, W. Thiel, and P. Jensen, J. Mol. Spectrosc. 245, 126 (2007).
5. S. N. Yurchenko, A. F. Al-Refaie and J. Tennyson, 'EXOCROSS: a general program for generating spectra from molecular line lists', A&A, 614 (2018) A131 DOI: https://doi.org/10.1051/0004-6361/201732531

ExoMol

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